

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Analytical Problem Solving

Code of Conduct:

I R.JAGAN-192321102(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

- (a) Explain the purpose of grey level transformation in image enhancement.
- (b) Apply a negative transformation $s=255-r$ to the pixel values: $r=[50,100,150,200]$
Compute the transformed values.

2. Log Transformation

- (a) Explain how log transformation enhances low-intensity pixels.
- (b) Given $s=c\log(1+r)$, where $c=10$, compute s for: $r=[0,10,50,100]$

3. Histogram Processing

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1
2

4
4

- (a) Explain the role of histogram equalization in contrast enhancement.
- (b) Given a 3-bit image with histogram: **Intensity Frequency**

Intensity	Frequency
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

- (a) Explain the purpose of spatial filtering for smoothing.
- (b) Apply a 3×3 averaging filter to the following
region: 10 20 30 20 30 40 30 40 50
Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

- (a) Explain how sharpening enhances image details.
- (b) Apply the Laplacian mask:
 $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ to the matrix:
10 10 10, 10 20 10, 10 10 10
Compute the output at the center.

ANSWERS:

1. Grey Level Transformation

(a) Purpose of Grey Level Transformation

Grey level transformation is an image enhancement technique that changes the intensity (brightness) values of pixels to improve the visual appearance of an image. It is used to:

- Enhance image contrast.
 - Highlight important features.
 - Brighten or darken images.
 - Prepare images for further processing.
- (b) Negative Transformation Formula:

$$s = 255 - r$$

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

Answer: [205, 155, 105, 55]

2. Log Transformation

(a) How Log Transformation Enhances Low-Intensity Pixels

The log transformation expands the values of dark (low-intensity) pixels while compressing the values of bright pixels. This makes details in dark regions more visible without excessively increasing the brightness of already bright areas. Formula:

Given: $s = c \log (1 + r)$

•

Using $c = \text{natural log base } 10$ ($\ln$):

r Calculation s 10

101010lnlnln (((101

(11511))))) 23.98

0		0.00
50		39.32
100	10	46.15

Answer: 0.00ln , 23.98, 39.32, 46.15

3. Histogram Processing

(a) Role of Histogram Equalization

Histogram equalization improves image contrast by redistributing the pixel intensity values over the full intensity range. It:

- Increases global contrast.
- Makes hidden details more visible.
- Produces a more uniform histogram.

(b) Cumulative Distribution and Equalized Intensities

Given histogram (3-bit image, $L = 8$):

Intensity Frequency $L = 8$

0 2

1 2

2 4

3 8

Total pixels:

Cumulative frequency: $N = 2 + 2 + 4 + 8 = 16$

Intensity Frequency Cumulative Frequency

0 2 2

1 2 4

2 4 8 3 16

Equalization formula:

$$s_k = \text{round} \left(\frac{CDF_N}{N} (L - 1) \right)$$

Intensity $L = 8$ $CDF_L - \text{Equalized} = 7$

Value where s_k is, so s_k .

0 2 1

1 4 2

2 8 4

3 16 7

Equalized Intensities:

Original Equalized

0 1

1 2

2 4

3 7

4. Spatial Filtering (Smoothing)

(a) Purpose of Spatial Filtering for Smoothing

Smoothing filters reduce noise and small intensity variations by replacing each pixel with the average of its neighboring pixels. They:

- Reduce image noise.
- Blur unwanted details.
- Produce a smoother image. (b) Apply a 3×3 Averaging Filter Matrix:

Sum of all values:

1
0
2
0
3
0
2
0
3
0
4
0
3
0
4
0
5
0

Average: $10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$

$$\frac{270}{9} = 30$$

New center pixel = 30

5. Spatial Filtering (Sharpening)

(a) How Sharpening Enhances Image Details

Sharpening enhances edges and fine details by emphasizing rapid intensity changes in an image. The Laplacian operator detects these intensity variations, making boundaries and textures appear sharper. (b) Apply the Laplacian Mask Mask:

Image: $[0 - 1 0 - 1 4 - 1 0 - 1 0]$

Calculation: $10 10 10 10 20 10 10 10 10$

Output at the center = 40 $(4 \times 20) - (10 + 10 + 10 + 10) = 80 - 40 = 40$